Automorphism groups from the link of a point

Finite application to six Higman–Sims geometries

Let $G = \mathrm{HS}$, with its acquired faithful permutation model, complete coset catalogue and identified group $N = \mathrm{Aut}(G)$. We retain the source coordinates of the supplement to D. Leemans, *The residually weakly primitive geometries of HS*, Australas. J. Combin. **33** (2005), 231–236. Our purpose is to determine the full groups in six specified geometries. The numerical classification, source coordinates and finite group interfaces are acquired premises. The reduction to a point link is proved here.

Lemma 1 (A point link bound)**.** *Let $G \le A \le \mathrm{Sym}(\Omega)$, where G is transitive on the finite set Ω. Suppose A preserves a family $\mathcal{B}$ of distinct subsets of Ω. For $b \in \Omega$, put*

$$\mathcal{L}_b = \{B \setminus \{b\} : B \in \mathcal{B},\ b \in B\}.$$

Let L be the group of permutations of $\Omega \setminus \{b\}$ preserving $\mathcal{L}_b$. Then $|A| \le |\Omega||L|$. In particular, if $|\Omega||L| = |G|$, then $A = G$.

Proof. Every element of A_b permutes the blocks containing b and therefore preserves their deletions. Its restriction to $\Omega \setminus \{b\}$ determines it, so A_b embeds in L. Since G is transitive, so is A, and orbit–stabilizer gives $|A| = |\Omega||A_b| \le |\Omega||L|$. The last assertion uses the actual containment $G \le A$. □

We compute L as the complete group of the bipartite incidence graph whose two parts are the points and the distinct members of $\mathcal{L}_b$. The parts have different colours. Distinct members have distinct point neighbourhoods, so an automorphism fixing every point fixes every block. Conversely a permutation preserving the family extends uniquely to this graph. Thus its full coloured graph group is exactly L; there is no unaccounted permutation kernel on repeated blocks.

Blocks determined by incidence. Choose a type a of cardinality 176, and let $\Omega = \{P_a g : g \in G\}$. For each other type j, take the distinct point shadows of its objects. At the base object the shadow is $S_j = b^{P_j}$, so these shadows form the orbit S_j^G and their number is $[G : G_{S_j}]$. Among types with $2 < |S_j| < 174$, choose the least triple

$$\big([G : G_{S_j}],\ [G : P_j],\ |S_j|\big)$$

in lexicographic order. Take the union of the shadow families of *all* types attaining this triple, discarding repeated subsets. This gives $\mathcal{B}$. Every type-preserving automorphism preserves it: the construction uses only incidences, cardinalities and distinct shadows. The group notation supplies a way to compute these intrinsic quantities.

Faithfulness of the restriction. The acquired shadow lemma supplies a sufficient condition for the full typed group to act faithfully on Ω. A type-j object's initial description is its point shadow. If its stabilizer in G equals P_j, the description is injective within that type. For a type not yet identified, append the set of descriptions of its neighbours of an already identified type k. At the base object this set is the P_j-orbit of the base description of type k, of size $[P_j : P_j \cap P_k]$. Intersecting the previous stabilizer with the stabilizer of this set gives the refined

stabilizer. Induction on these dependencies proves that a typed automorphism fixing Ω fixes every identified type. In all applications every final stabilizer equals the corresponding P_j. Only family 53 requires a refinement: type 1 uses its two type-2 neighbours.

Exact finite interfaces. For every anchor the saved images of the ordered generators give a checked bijection $\phi : G \longrightarrow E \leq \mathrm{Sym}(176)$. The verifier checks that the base point has stabilizer $\phi(P_a)$ and that the saved representatives carry it to every numbered point. These equations identify the action with the actual coset action. All point shadows, chosen triples and description stabilizers are then computed from the actual subgroup images.

The saved block list is checked without rediscovering its orbits. Every generator permutes the whole list, every selected base shadow is present, and a traversal from these seeds along generator edges reaches every listed block. Closure gives one inclusion in the required union; reachability gives the other. Since the action is finite, positive generator words suffice. This proves exact equality with $\mathcal{B}$. The link is formed directly from this verified list. Complete nauty output is an explicit computational premise. Each returned generator is also checked on every link block and on its unique extension to the graph.

Theorem 2. *In the six families below, the full type-preserving group equals the specified G. The full correlation groups are as shown. For each listed anchor the complete link group has order* 252 000.

Family	Source	Anchor	$(\lvert\mathcal{B}\rvert, \lvert B\rvert)$	$\mathrm{Cor}(\Gamma)$	$\lvert\mathcal{L}_b\rvert$
52	4.26	3,4	$(30800, 6)$	$G \times C_2^2$	1050
53	4.25	3,4	$(30800, 6)$	$G \times C_2$	1050
66	4.12	1	$(5775, 16)$	G	525
68	4.9	3	$(15400, 12)$	G	1050
		4	$(36960, 6)$		1260
69	4.10	3,4	$(36960, 6)$	$G \times C_2$	1260
70	4.8	4	$(36960, 6)$	G	1260

Proof. The faithful restriction, intrinsic block family and point link bound give $|\mathrm{Aut}(\Gamma)| \leq 176 \cdot 252000 = 44352000 = |G|$. The specified G already acts by typed automorphisms; hence equality holds as groups of automorphisms. In particular G is the kernel of the type-permutation action of $\mathrm{Cor}(\Gamma)$, and is normal there.

Put $B_0 = \bigcap_i P_i$. The acquired normalizer theorem therefore applies. If S, T are the ordered and setwise family stabilizers in the complete N, it gives

$$|\mathrm{Aut}(\Gamma)| = |G : B_0||S|, \qquad |\mathrm{Cor}(\Gamma)| = |G : B_0||T|,$$

and identifies the centralizer of G with $(T \cap G)/B_0$. Every type permutation is tested by complete successive subgroup transporters, and each positive lift is checked on all parabolics. The six values of $(|B_0|, |S|, |T|)$, in table order, are

$$(8, 8, 32), \quad (8, 8, 16), \quad (16, 16, 16), \quad (20, 20, 20), \quad (20, 20, 40), \quad (20, 20, 20).$$

The centralizer has respectively order $4, 2, 1, 1, 2, 1$; in the first case every element squares to the identity, proving that it is C_2^2. As G is centreless, its intersection with this centralizer is trivial. Their product has the computed full correlation order, which proves the direct products displayed in the table. □

Scope of the verification. All ten anchors are checked, with all four types identified in each. Damaged controls test closed extra orbits, omitted or duplicated blocks, wrong seed families, wrong actual subgroup data and missing faithfulness. Ordinary and optimized Python reports must agree. The computation repeats no ambient subgroup or geometry census. This application supplies six full-group determinations; it does not complete the source manuscript, settle an open conjecture or supply a Lean formalization.